

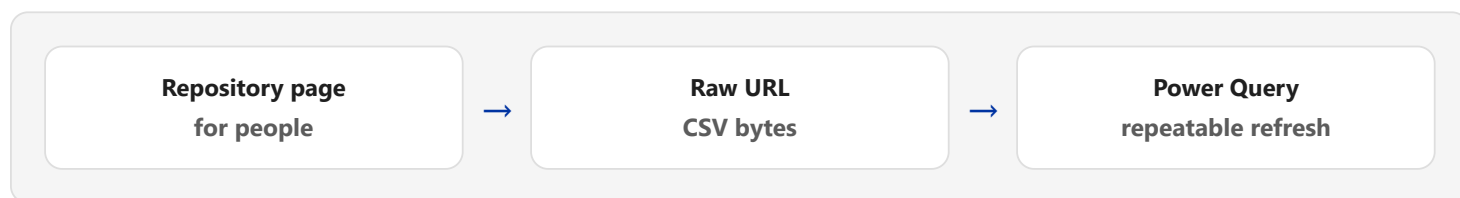
## LAB 1 • SOURCE ACQUISITION

# Connect Power BI to GitHub data

Use the Web connector against raw CSV URLs. You will stage three queries, name them consistently, and profile the source before changing it.

20–25 minutes Beginner Deliverable: 3 queries

## Why the raw URL matters



The GitHub **tree** URL returns an HTML page. Power BI needs the **raw.githubusercontent.com** URL, which returns only the CSV content.

## Source files

### Financial Data - Budget vs Actuals.csv

[https://raw.githubusercontent.com/Coding-Forge/PBI-Supply-Chain/main/Coding-Forge\\_Data/Financial%20Data%20-%20Budget%20vs%20Actuals.csv](https://raw.githubusercontent.com/Coding-Forge/PBI-Supply-Chain/main/Coding-Forge_Data/Financial%20Data%20-%20Budget%20vs%20Actuals.csv)

### Financial Data - Programs.csv

[https://raw.githubusercontent.com/Coding-Forge/PBI-Supply-Chain/main/Coding-Forge\\_Data/Financial%20Data%20-%20Programs.csv](https://raw.githubusercontent.com/Coding-Forge/PBI-Supply-Chain/main/Coding-Forge_Data/Financial%20Data%20-%20Programs.csv)

### Financial Data - Purchase Orders.csv

[https://raw.githubusercontent.com/Coding-Forge/PBI-Supply-Chain/main/Coding-Forge\\_Data/Financial%20Data%20-%20Purchase%20Orders.csv](https://raw.githubusercontent.com/Coding-Forge/PBI-Supply-Chain/main/Coding-Forge_Data/Financial%20Data%20-%20Purchase%20Orders.csv)

## Connect the first file

1

### Open a blank report.

In Power BI Desktop, select **Home** → **Get data**. Search for **web**, select **Web API**, and choose **Connect**.

2

### Paste the Budget vs Actuals raw URL.

Paste the first raw URL above into the **URL** field.

3

### Configure public access.

Set **Authentication kind** to **Anonymous**. For this public training repository, leave **Privacy Level** at **None**, then choose **Next**.

4

### Open Power Query.

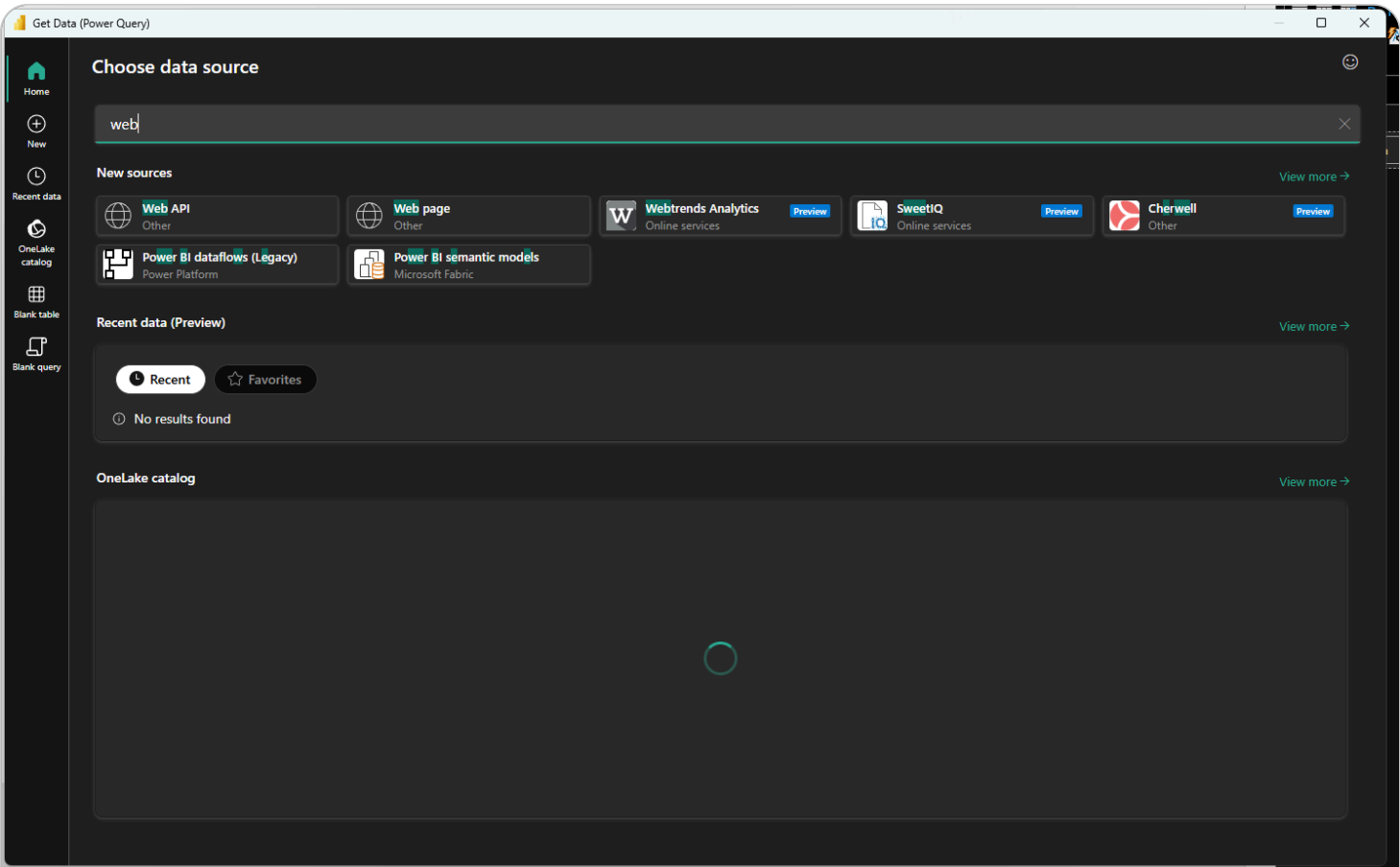
Confirm the preview contains 13 columns. Open the arrow beside **Load** and select **Transform data**.

5

### Rename the query.

In Query Settings, change the name to **Budget vs Actuals**. Remove an automatic **Changed Type** step if it created errors; types are handled in Lab 2.

# Import walkthrough



## 1. Choose Web API

Search for "web," then select the Web API connector. Power BI labels may vary slightly by Desktop release.

Get Data (Power Query)

### Connect to data source

**Web API**  
Other  
[learn more](#)

#### Connection settings

URL \*

#### Connection credentials

Authentication kind

Anonymous

Privacy Level

None

Back

Cancel

Next

## 2. Configure the connection

Paste the raw GitHub URL, use Anonymous authentication, and leave Privacy Level at None for this public source.

Get Data (Power Query)

Preview file data

URL: https://raw.githubusercontent.com/Coding-Forge/PBI-Supply-Chain/main/Coding-Forge\_Data/Financial%20Data%20-%20Budget%20vs%20Actuals.csv

File origin

65001: Unicode (UTF-8)

Delimiter

Comma

Data type detection

Based on first 200 rows

| PRG- Program ID | PRG- Program Name                | Fiscal Year | Quarter | Cost Center | Budget Category | Budgeted Amount | Actual Amount | Variance      | Variance % | Funding Source    | Contract Number  | Status      |
|-----------------|----------------------------------|-------------|---------|-------------|-----------------|-----------------|---------------|---------------|------------|-------------------|------------------|-------------|
| PRG-001         | Thunderbolt Sustainment Support  | FY2025      | Q1      | CC-4100     | Labor           | 2,500,000.00    | \$2,350,000   | \$150,000     | 6.00%      | DoD Appropriation | FA8615-24-C-0012 | On Track    |
| PRG-001         | Thunderbolt Sustainment Support  | FY2025      | Q1      | CC-4100     | materials       | 1,800,000.00    | \$1,920,000   | (\$120,000)   | -6.70%     | DoD Appropriation | FA8615-24-C-0012 | over budget |
| PRG-001         | Thunderbolt Sustainment Support  | FY2025      | Q1      | CC-4100     | Travel          | 350,000.00      | \$280,000     | \$70,000      | 20.00%     | DoD Appropriation | FA8615-24-C-0012 | On Track    |
| PRG-001         | Thunderbolt Sustainment Support  | FY2025      | Q1      | CC-4100     | Sub-Contracts   | 4,200,000.00    | \$4,150,000   | \$50,000      | 1.20%      | DoD Appropriation | FA8615-24-C-0012 | On Track    |
| PRG-001         | Thunderbolt Sustainment Support  | FY2025      | Q2      | CC-4100     | Labor           | 2,500,000.00    | \$3,900,000   | (\$1,400,000) | -56.00%    | DoD Appropriation | FA8615-24-C-0012 | Over Budget |
| PRG-001         | Thunderbolt Sustainment Support  | FY2025      | Q2      | CC-4100     | Materials       | 1,800,000.00    | \$1,750,000   | \$50,000      | 2.80%      | DoD Appropriation | FA8615-24-C-0012 | On Track    |
| PRG-001         | Thunderbolt Sustainment Support  | FY2025      | Q2      | CC-4100     | travel          | 350,000.00      | \$580,000     | (\$230,000)   | -65.70%    | DoD Appropriation | FA8615-24-C-0012 | Over Budget |
| PRG-001         | Thunderbolt Sustainment Support  | FY2025      | Q2      | CC-4100     | Subcontracts    | 4,200,000.00    | \$5,800,000   | (\$1,600,000) | -38.10%    | DoD Appropriation | FA8615-24-C-0012 | OVER BUDGET |
| PRG-002         | Cyber Defense Initiative         | FY2025      | Q1      | CC-5200     | Labor           | 1,200,000.00    | \$750,000     | \$450,000     | 37.50%     | DARPA Grant       | W31P4Q-24-C-0088 | On Track    |
| PRG-002         | Cyber Defense Initiative         | FY2025      | Q1      | CC-5200     | Materials       | 400,000.00      | \$380,000     | \$20,000      | 5.00%      | DARPA grant       | W31P4Q-24-C-0088 | On Track    |
| PRG-002         | Cyber Defense Initiative         | FY2025      | Q1      | CC-5200     | Travel          | 150,000.00      | \$145,000     | \$5,000       | 3.30%      | DARPA Grant       | W31P4Q-24-C-0088 | On Track    |
| PRG-002         | Cyber Defense Initiative         | FY2025      | Q1      | CC-5200     | Sub-Contracts   | 800,000.00      | \$790,000     | \$10,000      | 1.30%      | DARPA Grant       | W31P4Q-24-C-0088 | On Track    |
| PRG-002         | Cyber Defense Initiative         | FY2025      | Q2      | CC-5200     | Labor           | 1,200,000.00    | \$780,000     | \$420,000     | 35.00%     | DARPA Grant       | W31P4Q-24-C-0088 | On Track    |
| PRG-002         | Cyber Defense Initiative         | FY2025      | Q2      | CC-5200     | Materials       | 400,000.00      | \$420,000     | (\$20,000)    | -5.00%     | DARPA Grant       | W31P4Q-24-C-0088 | At Risk     |
| PRG-002         | Cyber Defense Initiative         | FY2025      | Q2      | CC-5200     | Travel          | 150,000.00      | \$130,000     | \$20,000      | 13.30%     | DARPA Grant       | W31P4Q-24-C-0088 | on track    |
| PRG-002         | Cyber Defense Initiative         | FY2025      | Q2      | CC-5200     | Subcontracts    | 800,000.00      | \$810,000     | (\$10,000)    | -1.30%     | DARPA Grant       | W31P4Q-24-C-0088 | On Track    |
| PRG-003         | Satellite Comm Upgrade           | FY2025      | Q1      | CC-3300     | Labor           | 3,100,000.00    | \$3,050,000   | \$50,000      | 1.60%      | Space Force       | FA8808-24-C-0045 | On Track    |
| PRG-003         | Satellite Comm Upgrade           | FY2025      | Q1      | CC-3300     | Materials       | 2,200,000.00    | \$2,400,000   | (\$200,000)   | -9.10%     | Space Force       | FA8808-24-C-0045 | Over Budget |
| PRG-003         | Satellite Comm Upgrade           | FY2025      | Q1      | CC-3300     | Travel          | 200,000.00      | \$180,000     | \$20,000      | 10.00%     | Space Force       | FA8808-24-C-0045 | On Track    |
| PRG-003         | Satellite Communications Upgrade | FY2025      | Q1      | CC-3300     | Sub-Contracts   | 5,500,000.00    | \$7,400,000   | (\$1,900,000) | -34.50%    | Space Force       | FA8808-24-C-0045 | Over Budget |
| PRG-003         | Satellite Comm Upgrade           | FY2025      | Q2      | CC-3300     | Labor           | 3,100,000.00    | \$3,200,000   | (\$100,000)   | -3.20%     | Space Force       | FA8808-24-C-0045 | At Risk     |
| PRG-003         | Satellite Comm Upgrade           | FY2025      | Q2      | CC-3300     | Materials       | 2,200,000.00    | \$2,100,000   | \$100,000     | 4.50%      | Space Force       | FA8808-24-C-0045 | On Track    |
| PRG-003         | Satellite Comm Upgrade           | FY2025      | Q2      | CC-3300     | Travel          | 200,000.00      | \$480,000     | (\$280,000)   | -140.00%   | Space Force       | FA8808-24-C-0045 | OVER BUDGET |
| PRG-003         | Satellite Comm Upgrade           | FY2025      | Q2      | CC-3300     | Subcontracts    | 5,500,000.00    | \$7,500,000   | (\$2,000,000) | -36.40%    | Space Force       | FA8808-24-C-0045 | OVER BUDGET |
| PRG-004         | Logistics Command Platform       | FY2025      | Q1      | CC-6100     | Labor           | 900,000.00      | \$580,000     | \$320,000     | 35.60%     | Army RDECOM       | W56HZV-24-C-0199 | On Track    |
| PRG-004         | Logistics Command Platform       | FY2025      | Q1      | CC-6100     | Materials       | 300,000.00      | \$290,000     | \$10,000      | 3.30%      | Army RDECOM       | W56HZV-24-C-0199 | On Track    |
| PRG-004         | Logistics Command Platform       | FY2025      | Q1      | CC-6100     | Travel          | 100,000.00      | \$95,000      | \$5,000       | 5.00%      | Army RDECOM       | W56HZV-24-C-0199 | On Track    |
| PRG-004         | Logistics Command Platform       | FY2025      | Q1      | CC-6100     | Sub-Contracts   | 600,000.00      | \$580,000     | \$20,000      | 3.30%      | Army RDECOM       | W56HZV-24-C-0199 | On Track    |
| PRG-004         | Logistics Command Platform       | FY2025      | Q2      | CC-6100     | Labor           | 900,000.00      | \$350,000     | \$350,000     | 38.90%     | Army RDECOM       | W56HZV-24-C-0199 | On Track    |
| PRG-004         | Logistics Command Platform       | FY2025      | Q2      | CC-6100     | Materials       | 300,000.00      | \$310,000     | (\$10,000)    | -3.30%     | Army RDECOM       | W56HZV-24-C-0199 | on track    |
| PRG-004         | Logistics Command Platform       | FY2025      | Q2      | CC-6100     | Travel          | 100,000.00      | N/A           | N/A           | N/A        | Army RDECOM       | W56HZV-24-C-0199 |             |

Load

Transform data

Back

Cancel

Add table using examples

Load

### 3. Inspect before transforming

Verify that all 13 columns appear, then choose Transform data from the Load menu to continue in Power Query.

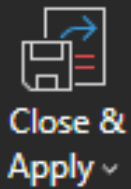
Home

Transform

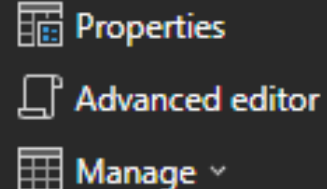
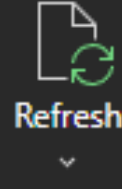
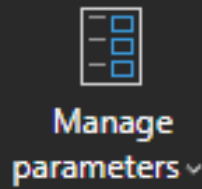
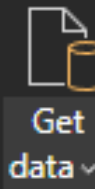
Add column

View

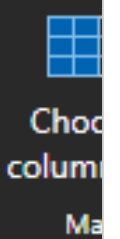
Help



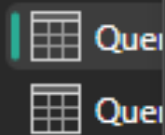
Close



Query



Queries



Excel workbook



SQL Server database



Text/CSV



Web page

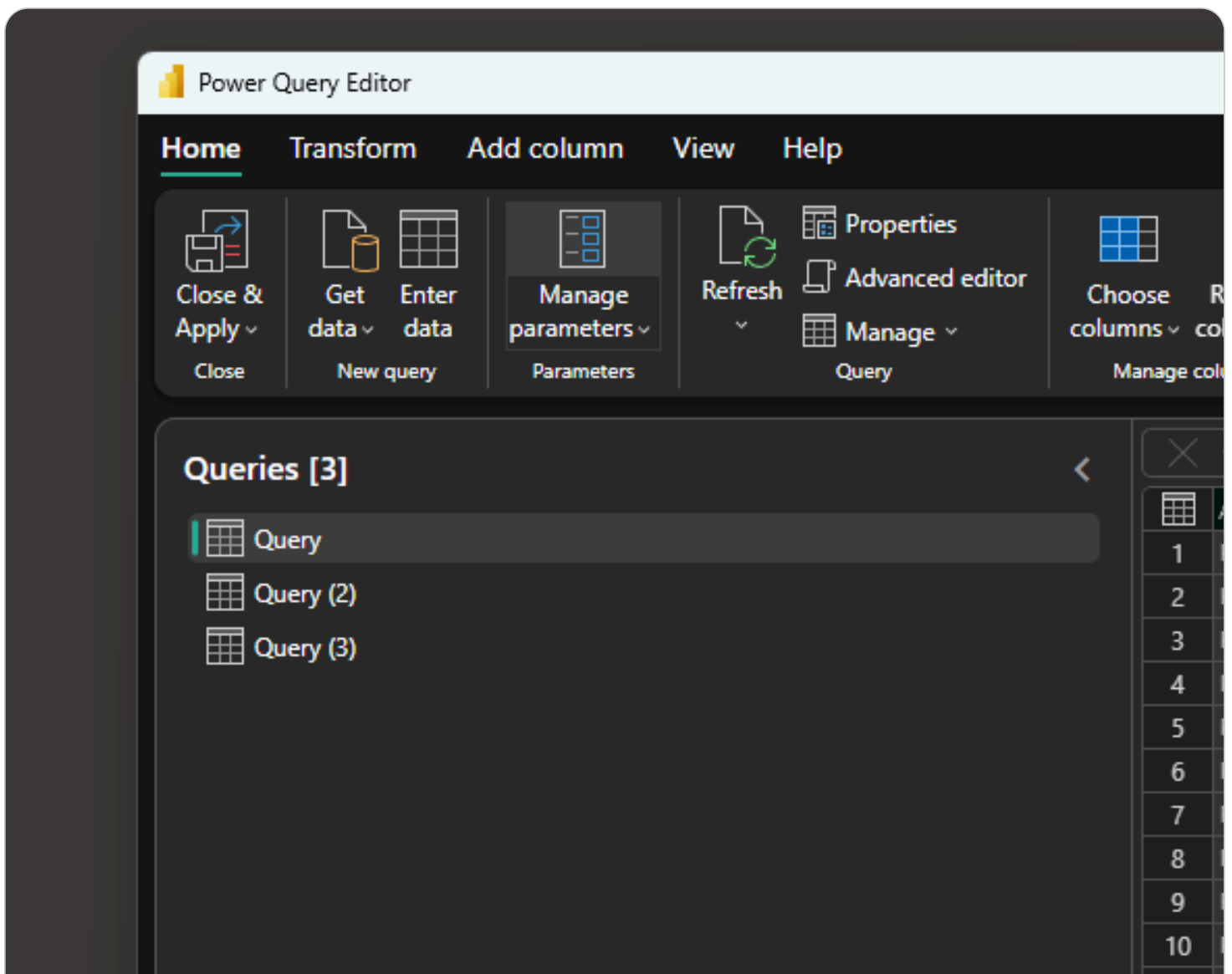


Blank query

More...

#### 4. Find additional connectors

Inside Power Query Editor, choose Get data → More... to open the complete connector catalog. Use this route to add the Programs and Purchase Orders Web API sources.



### 5. Rename the imported queries

Double-click each name in the Queries pane and assign the descriptive names Budget vs Actuals, Program, and Purchase Orders.

## Add the other two files

1

#### Open the complete connector list.

In Power Query Editor, select **Home** → **Get data** → **More....** Search for **web**, select **Web API**, and paste the Programs.csv URL.

2

#### Repeat for Purchase Orders.

Open **Get data** → **More...** again, select Web API, and paste the third raw URL. The source has 17 columns and 21 rows, including one intentional duplicate.

3

### Rename all three queries.

In the Queries pane, double-click each default name and use **Budget vs Actuals**, **Program**, and **Purchase Orders**. Program should show 4 columns and 5 data rows.

4

### Save the report.

Use **File** → **Save** and name it **Coding-Forge-Supply-Chain-Lab.pbix**. Stay in Power Query for Lab 2.

## Profile before cleaning

On the **View** tab, enable **Column quality**, **Column distribution**, and **Column profile**. Set profiling to **Based on entire data set**.

Find at least one example of each:

- Currency stored as text and negative values in parentheses.
- N/A, NULL, or - standing in for missing values.
- Mixed capitalization and extra spaces.
- Mixed date formats.
- A duplicate purchase order and footer rows in Budget vs Actuals.

## Recovery options

**Web access blocked?** Download the three files from the repository's **Coding-Forge\_Data** folder, then use **Get data** → **Text/CSV**. Keep the filenames exactly as listed above.

**Privacy-level warning?** Edit permissions under **File** → **Options and settings** → **Data source settings**. Use Anonymous credentials and Public privacy for this public training repository.